

EXTENSION 1

Stage 6

Calculus (Ext1), C3 Applications of Calculus (Y12)

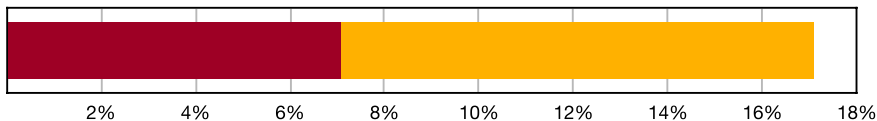
Further Area and Solids of Revolution (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 180 minutes (based on allocation of 1.5 minutes per mark)



C3 Applications of Calculus



Further Area and Solids of Revolution
Differential Equations

*SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.

HISTORICAL CONTRIBUTION

- C3 Application of Calculus* is the 880-pound Silverback of the Extension 1 course, contributing a huge 17.1% per new syllabus Ext1 exam since the new course began in 2020.
- This topic has been split into two sub-categories for analysis purposes which are: *1-Further Areas and Solids of Revolution (7.1%)* and *2-Differential Equations (10.0%)*.
- This analysis looks at *Further Areas and Solids of Revolution*.

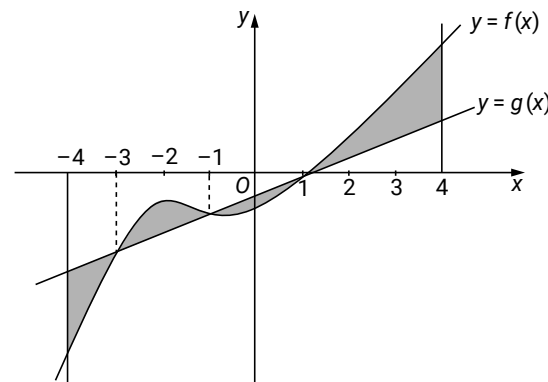
HSC ANALYSIS - What to expect and common pitfalls

- Further Areas and Solids of Revolution (7.1%)* is a sneaky large contributor, having been tested within the longer answer section every year, and attracting two questions in three of those years.
- The smaller topic of *Further Area* has been examined in 3 separate exams and is an important revision area (see *2021 Ext1 13c* and *2024 Ext1 11g*).
- Solids of Revolution* questions are typically examined around the band 4 difficulty level. *2023 Ext1 12e* is the exception, raising the difficulty to Band 5 and producing a 49% mean mark - a revision focus question.
- Note the database is a combination of Ext1 exam questions and others taken from past *Advanced* exams which can be identified by EXT1* in their title.
- History shows students have more difficulty with *y*-axis rotations. Ensure this question-type is carefully reviewed.

Questions

1. Calculus, EXT1 C3 2024 HSC 2 MC

Consider the functions $y = f(x)$ and $y = g(x)$, and the regions shaded in the diagram below.

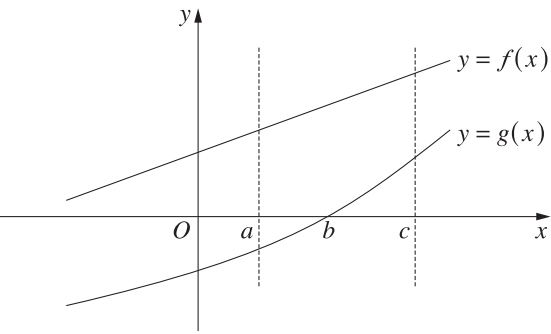


Which of the following gives the total area of the shaded regions?

- A. $\int_{-4}^4 f(x) - g(x) dx$
- B. $\left| \int_{-4}^4 f(x) - g(x) dx \right|$
- C. $\int_{-4}^{-3} f(x) - g(x) dx + \int_{-3}^{-1} f(x) - g(x) dx + \int_{-1}^1 f(x) - g(x) dx + \int_1^4 f(x) - g(x) dx$
- D. $-\int_{-4}^{-3} f(x) - g(x) dx + \int_{-3}^{-1} f(x) - g(x) dx - \int_{-1}^1 f(x) - g(x) dx + \int_1^4 f(x) - g(x) dx$

2. Calculus, EXT1 C3 2023 HSC 4 MC

The diagram shows the graphs of the functions $f(x)$ and $g(x)$.



It is known that

$$\int_a^c f(x)dx = 10$$

$$\int_a^b g(x)dx = -2$$

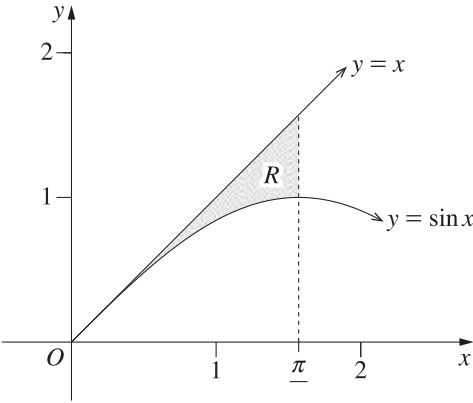
$$\int_b^c g(x)dx = 3.$$

What is the area between the curves $y = f(x)$ and $y = g(x)$ between $x = a$ and $x = c$?

- A. 5
- B. 7
- C. 9
- D. 11

3. Calculus, EXT1 C3 2024 HSC 11g

The region, R , is bounded by the curves $y = \sin x$, $y = x$ and the line $x = \frac{\pi}{2}$ as shown in the diagram.



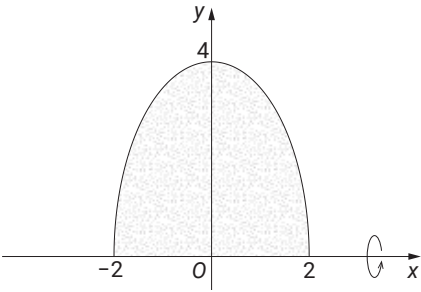
Find the area of the region R . (3 marks)

4. Calculus, EXT1 C3 2024 HSC 12b

The region, R , is bounded by the function, $y = x^3$, the x -axis and the lines $x = 1$ and $x = 2$.
What is the volume of the solid of revolution obtained when the region R is rotated about the x -axis? (3 marks)

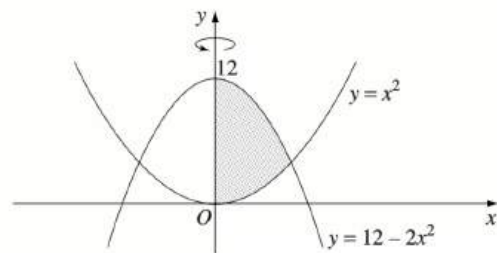
5. Calculus, EXT1* C3 2017 HSC 12b

The diagram shows the region bounded by $y = \sqrt{16 - 4x^2}$ and the x -axis.



The region is rotated about the x -axis to form a solid.
Find the exact volume of the solid formed. (3 marks)

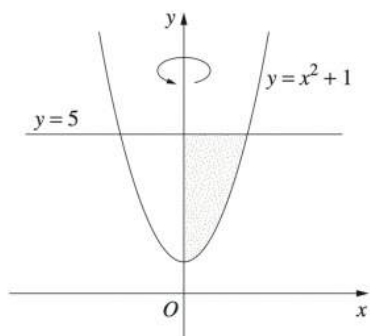
6. Calculus, EXT1* C3 2005 HSC 6c



The graphs of the curves $y = x^2$ and $y = 12 - 2x^2$ are shown in the diagram.

- Find the points of intersection of the two curves. **(1 mark)**
- The shaded region between the curves and the y -axis is rotated about the y -axis. By splitting the shaded region into two parts, or otherwise, find the volume of the solid formed. **(3 marks)**

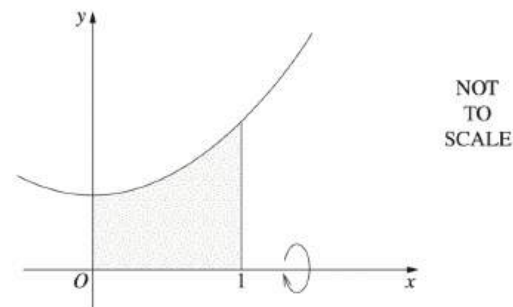
7. Calculus, EXT1* C3 2006 HSC 4b



In the diagram, the shaded region is bounded by the parabola $y = x^2 + 1$, the y -axis and the line $y = 5$.

Find the volume of the solid formed when the shaded region is rotated about the y -axis. **(3 marks)**

8. Calculus, EXT1* C3 2007 HSC 9a



The shaded region in the diagram is bounded by the curve $y = x^2 + 1$, the x -axis, and the lines $x = 0$ and $x = 1$.

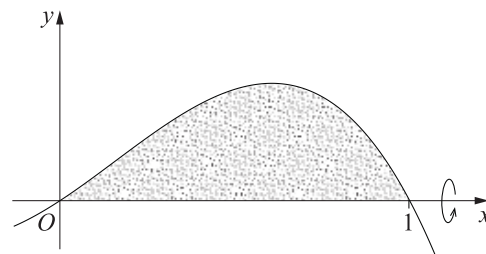
Find the volume of the solid of revolution formed when the shaded region is rotated about the x -axis. **(3 marks)**

9. Calculus, EXT1 C3 EQ-Bank 1

- Sketch the region bounded by the curve $y = x^2$ and the lines $y = 16$ and $y = 9$. **(1 mark)**
- Calculate the area of this region. **(3 marks)**

10. Calculus, EXT1* C3 2019 HSC 13d

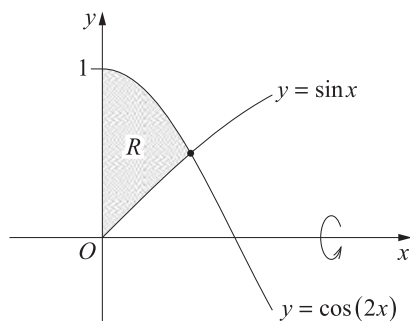
The diagram shows the region bounded by the curve $y = x - x^3$, and the x -axis between $x = 0$ and $x = 1$. The region is rotated about the x -axis to form a solid.



Find the exact value of the volume of the solid formed. **(3 marks)**

11. Calculus, EXT1 C3 2020 HSC 13b

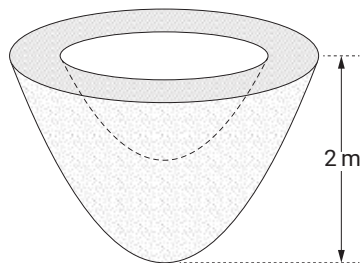
The region R is bounded by the y -axis, the graph of $y = \cos(2x)$ and the graph of $y = \sin x$, as shown in the diagram.



Find the volume of the solid of revolution formed when the region R is rotated about the x -axis. (4 marks)

12. Calculus, EXT1 C3 2021 HSC 13a

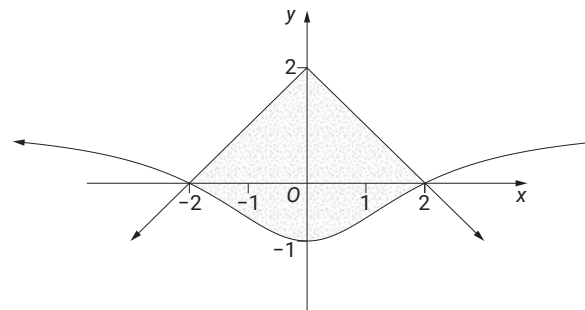
A 2-metre-high sculpture is to be made out of concrete. The sculpture is formed by rotating the region between $y = x^2$, $y = x^2 + 1$ and $y = 2$ around the y -axis.



Find the volume of concrete needed to make the sculpture. (3 marks)

13. Calculus, EXT1 C3 2021 HSC 13c

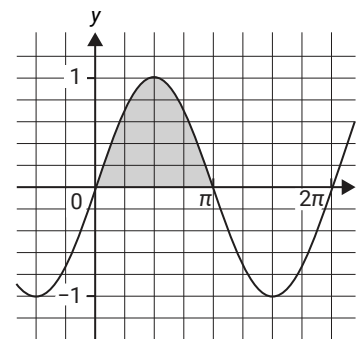
The region enclosed by $y = 2 - |x|$ and $y = 1 - \frac{8}{4 + x^2}$ is shaded in the diagram.



Find the exact value of the area of the shaded region. (3 marks)

14. Calculus, EXT1 C3 2021 SPEC1 4

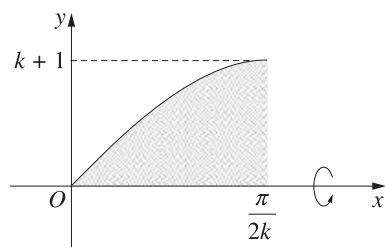
The shaded region in the diagram below is bounded by the graph of $y = \sin(x)$ and the x -axis between the first two non-negative x -intercepts of the curve, that is interval $[0, \pi]$. The shaded region is rotated about the x -axis to form a solid of revolution.



Find the volume, V_s , of the solid formed. (3 marks)

15. Calculus, EXT1 C3 2022 HSC 13b

A solid of revolution is to be found by rotating the region bounded by the x -axis and the curve $y = (k + 1)\sin(kx)$, where $k > 0$, between $x = 0$ and $x = \frac{\pi}{2k}$ about the x -axis.



Find the value of k for which the volume is π^2 . (3 marks)

16. Calculus, EXT1 C3 SM-Bank 2

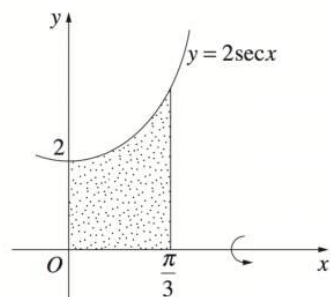
The parabola with equation $y = 9 - x^2$ cuts the y -axis at $P(0, 9)$ and the x -axis at $Q(3, 0)$.

Find the exact volume of the solid of revolution formed when the area between the line PQ and the parabola is rotated about the y -axis. (4 marks)

17. Calculus, EXT1 C3 SM-Bank 3

Find the volume of the solid of revolution formed when the graph of $y = \sqrt{\frac{1+2x}{1+x^2}}$ is rotated about the x -axis over the interval $[0, 1]$. (3 marks)

18. Calculus, EXT1* C3 2004 HSC 4c

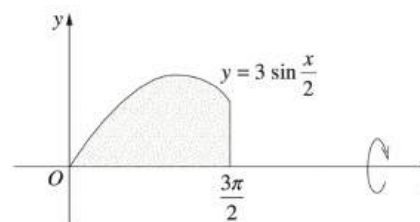


In the diagram, the shaded region is bounded by the curve $y = 2\sec x$, the coordinate axes and the line $x = \frac{\pi}{3}$. The shaded region is rotated about the x -axis.

Calculate the exact volume of the solid of revolution formed. (3 marks)

19. Calculus, EXT1 C3 2013 HSC 12b

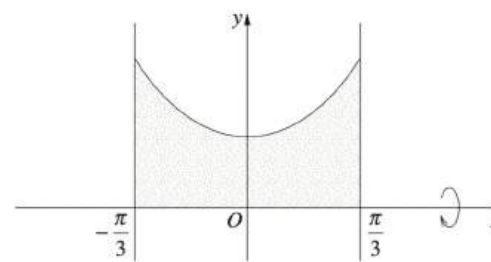
The region bounded by the graph $y = 3\sin \frac{x}{2}$ and the x -axis between $x = 0$ and $x = \frac{3\pi}{2}$ is rotated about the x -axis to form a solid.



Find the exact volume of the solid. (3 marks)

20. Calculus, EXT1* C3 2009 HSC 6a

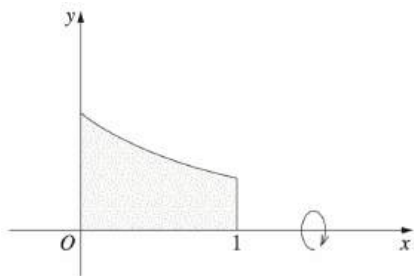
The diagram shows the region bounded by the curve $y = \sec x$, the lines $x = \frac{\pi}{3}$ and $x = -\frac{\pi}{3}$, and the x -axis.



The region is rotated about the x -axis. Find the volume of the solid of revolution formed. (3 marks)

21. Calculus, EXT1* C3 2012 HSC 14b

The diagram shows the region bounded by $y = \frac{3}{(x+2)^2}$, the x -axis, the y -axis, and the line $x = 1$.

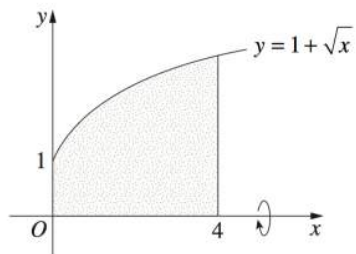


The region is rotated about the x -axis to form a solid.

Find the volume of the solid. (3 marks)

22. Calculus, EXT1* C3 2014 HSC 14c

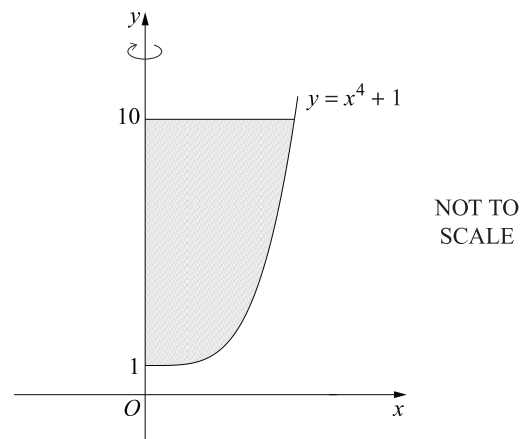
The region bounded by the curve $y = 1 + \sqrt{x}$ and the x -axis between $x = 0$ and $x = 4$ is rotated about the x -axis to form a solid.



Find the volume of the solid. (3 marks)

23. Calculus, EXT1* C3 2018 HSC 14b

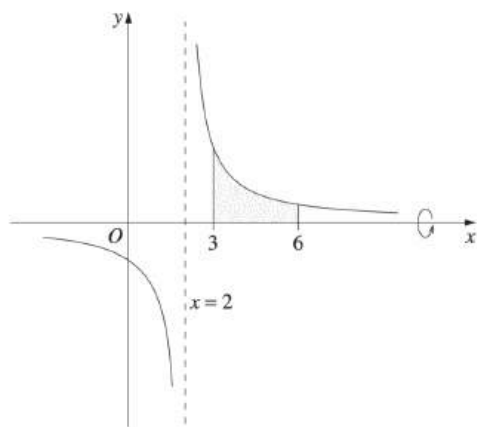
The shaded region shown in the diagram is bounded by the curve $y = x^4 + 1$, the y -axis and the line $y = 10$.



Find the volume of the solid of revolution formed when the shaded region is rotated about the y -axis. (3 marks)

24. Calculus, EXT1* C3 2008 HSC 6c

The graph of $y = \frac{5}{x-2}$ is shown below.



The shaded region in the diagram is bounded by the curve $y = \frac{5}{x-2}$, the x -axis and the lines $x = 3$ and $x = 6$.

Find the volume of the solid of revolution formed when the shaded region is rotated about the x -axis. **(3 marks)**

25. Calculus, EXT1 C3 2005 HSC 5a

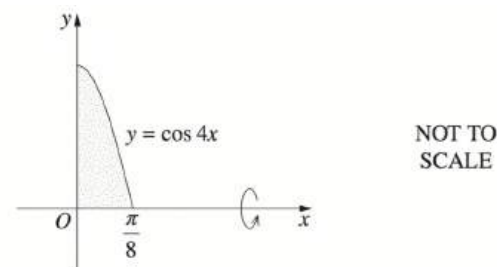
Find the exact value of the volume of the solid of revolution formed when the region bounded by the curve $y = \sin 2x$, the x -axis and the line $x = \frac{\pi}{8}$ is rotated about the x -axis. **(3 marks)**

26. Calculus, EXT1* C3 2007 HSC 3a

Find the volume of the solid of revolution formed when the region bounded by the curve $y = \frac{1}{\sqrt{9+x^2}}$, the x -axis, the y -axis and the line $x = 3$, is rotated about the x -axis. **(3 marks)**

27. Calculus, EXT1 C3 2014 HSC 12b

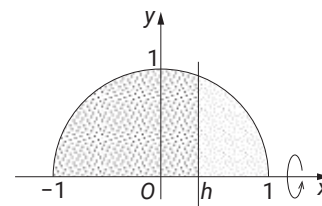
The region bounded by $y = \cos 4x$ and the x -axis, between $x = 0$ and $x = \frac{\pi}{8}$, is rotated about the x -axis to form a solid.



Find the volume of the solid. **(3 marks)**

28. Calculus, EXT1 C3 SM-Bank 1

The region enclosed by the semicircle $y = \sqrt{1-x^2}$ and the x -axis is to be divided into two pieces by the line $x = h$, when $0 \leq h < 1$.



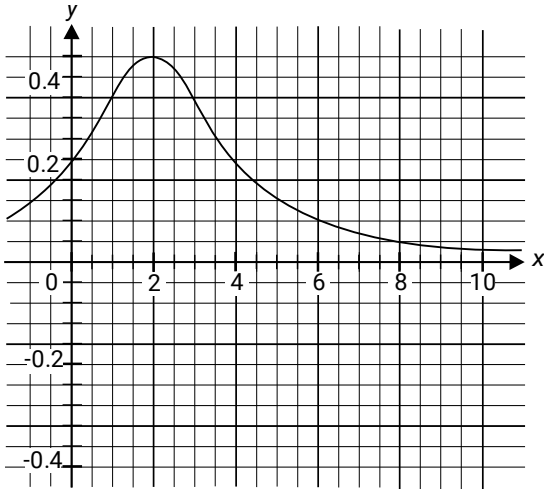
The two pieces are rotated about the x -axis to form solids of revolution. The value of h is chosen so that the volumes of the solids are in the ratio 2 : 1.

Show that h satisfies the equation $3h^3 - 9h + 2 = 0$. **(3 marks)**

29. Calculus, EXT1 C3 2019 SPEC1-N 9

i. Show that $\tan\left(\frac{5\pi}{12}\right) = \sqrt{3} + 2$. (2 marks)

ii.

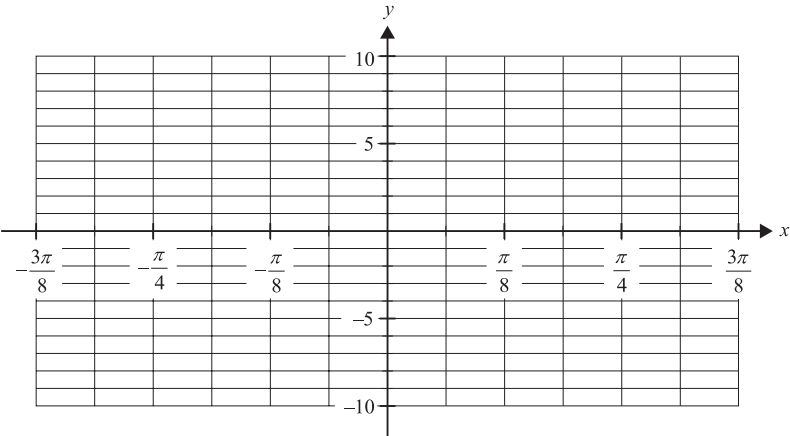


Hence, find the area bounded by the graph of $f(x) = \frac{2}{x^2 - 4x + 8}$ shown above, the x -axis and the lines $x = 0$ and $x = 2\sqrt{3} + 6$. (4 marks)

30. Calculus, EXT1 C3 2022 SPEC1 10

Let $f(x) = \sec(4x)$.

a. Sketch the graph of f for $x \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ on the set of axes below. Label any asymptotes with their equations and label any turning points and the endpoints with their coordinates. (3 marks)

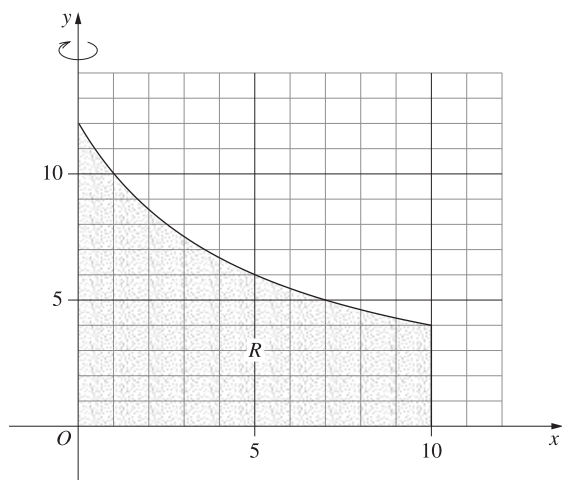


b. The graph of $y = f(x)$ for $x \in \left[-\frac{\pi}{24}, \frac{\pi}{48}\right]$ is rotated about the x -axis to form a solid of revolution.

Find the volume of this solid. Give your answer in the form $\frac{(a - \sqrt{b})\pi}{c}$, where $a, b, c \in \mathbb{R}$. (3 marks)

31. Calculus, EXT1 C3 2023 HSC 12e

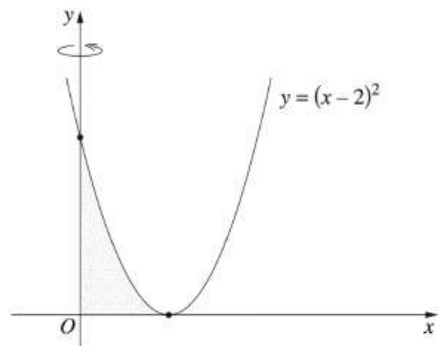
The region, R , bounded by the hyperbola $y = \frac{60}{x+5}$, the line $x = 10$ and the coordinate axes is shown.



Find the volume of the solid of revolution formed when the region R is rotated about the y -axis. Leave your answer in exact form. (4 marks)

32. Calculus, EXT1* C3 2013 HSC 15b

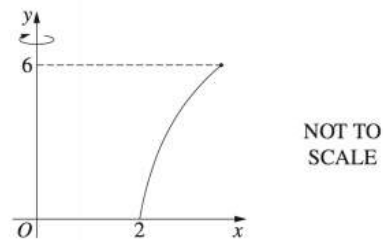
The region bounded by the x -axis, the y -axis and the parabola $y = (x-2)^2$ is rotated about the y -axis to form a solid.



Find the volume of the solid. (4 marks)

33. Calculus, EXT1* C3 2015 HSC 16b

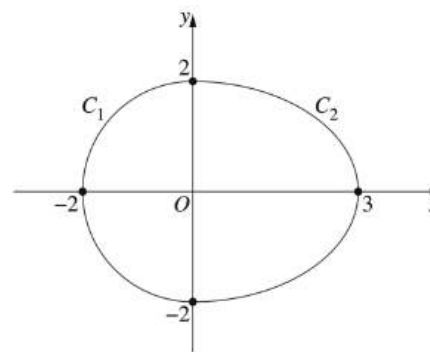
A bowl is formed by rotating the curve $y = 8\log_e(x-1)$ about the y -axis for $0 \leq y \leq 6$.



Find the volume of the bowl. Give your answer correct to 1 decimal place. (3 marks)

34. Calculus, EXT1* C3 2016 HSC 15a

The diagram shows two curves C_1 and C_2 . The curve C_1 is the semicircle $x^2 + y^2 = 4$, $-2 \leq x \leq 0$. The curve C_2 has equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$, $0 \leq x \leq 3$.



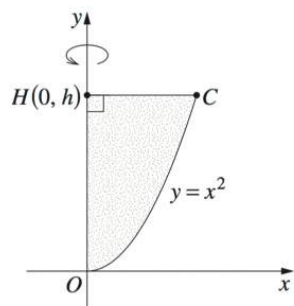
An egg is modelled by rotating the curves about the x -axis to form a solid of revolution.

Find the exact value of the volume of the solid of revolution. (4 marks)

35. Calculus, EXT1* C3 2011 HSC 8b

The diagram shows the region enclosed by the parabola $y = x^2$, the y -axis and the line $y = h$, where $h > 0$. This region is rotated about the y -axis to form a solid called a paraboloid. The point C is the intersection of $y = x^2$ and $y = h$.

The point H has coordinates $(0, h)$.



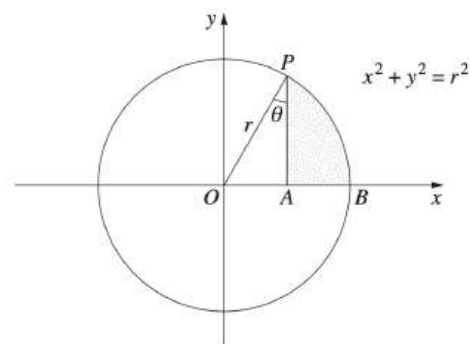
i. Find the exact volume of the paraboloid in terms of h . **(2 marks)**

ii. A cylinder has radius HC and height h .

What is the ratio of the volume of the paraboloid to the volume of the cylinder? **(1 mark)**

36. Calculus, EXT1* C3 2010 HSC 10b

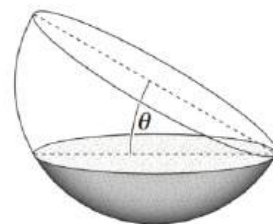
The circle $x^2 + y^2 = r^2$ has radius r and centre O . The circle meets the positive x -axis at B . The point A is on the interval OB . A vertical line through A meets the circle at P . Let $\theta = \angle OPA$.



i. The shaded region bounded by the arc PB and the intervals AB and AP is rotated about the x -axis. Show that the volume, V , formed is given by

$$V = \frac{\pi r^3}{3} (2 - 3\sin\theta + \sin^3\theta) \quad \textbf{(3 marks)}$$

ii. A container is in the shape of a hemisphere of radius r metres. The container is initially horizontal and full of water. The container is then tilted at an angle of θ to the horizontal so that some water spills out.



(1) Find θ so that the depth of water remaining is one half of the original depth. **(1 mark)**

(2) What fraction of the original volume is left in the container? **(2 marks)**

Worked Solutions

1. Calculus, EXT1 C3 2024 HSC 2 MC

Intervals where $f(x) > g(x) \Rightarrow$ Positive area values

Intervals where $g(x) > f(x) \Rightarrow$ Negative area values

$\Rightarrow D$

2. Calculus, EXT1 C3 2023 HSC 4 MC

$$A = \int_a^c f(x) dx - \int_b^c g(x) dx + \left| \int_a^b g(x) dx \right|$$

$$= 10 - 3 + |-2|$$

$$= 9$$

$\Rightarrow C$

3. Calculus, EXT1 C3 2024 HSC 11g

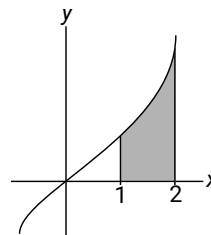
$$R = \int_0^{\frac{\pi}{2}} x - \sin x dx$$

$$= \left[\frac{x^2}{2} + \cos x \right]_0^{\frac{\pi}{2}}$$

$$= \left[\left(\frac{\pi^2}{8} + \cos \frac{\pi}{2} \right) - (0 + \cos 0) \right]$$

$$= \frac{\pi^2}{8} - 1 \text{ u}^2$$

4. Calculus, EXT1 C3 2024 HSC 12b



$$V = \pi \int_1^2 y^2 dx$$

$$= \pi \int_1^2 x^6 dx$$

$$= \pi \left[\frac{x^7}{7} \right]_1^2$$

$$= \pi \left(\frac{2^7 - 1}{7} \right)$$

$$= \frac{127\pi}{7} \text{ u}^3$$

5. Calculus, EXT1* C3 2017 HSC 12b

$$y = \sqrt{16-4x^2}$$

$$V = \pi \int_{-2}^2 y^2 dx$$

$$= 2\pi \int_0^2 16-4x^2 dx$$

$$= 2\pi \left[16x - \frac{4}{3}x^3 \right]_0^2$$

$$= 2\pi \left[\left(16 \cdot 2 - \frac{4}{3}2^3 \right) - 0 \right]$$

$$= 2\pi \left(\frac{64}{3} \right)$$

$$= \frac{128\pi}{3} \text{ u}^3$$

6. Calculus, EXT1* C3 2005 HSC 6c

i. $y = x^2 \qquad \dots (1)$

$y = 12 - 2x^2 \dots (2)$

Substitute $y = x^2$ into (2)

$$x^2 = 12 - 2x^2$$

$$3x^2 - 12 = 0$$

$$3(x^2 - 4) = 0$$

$$x = \pm 2$$

When $x = 2,$ $y = 4$

When $x = -2,$ $y = 4$

$\therefore$ Intersection at $(2, 4), (-2, 4)$

ii. In (1), $x^2 = y$

In (2), $y = 12 - 2x^2$

$$2x^2 = 12 - y$$

$$x^2 = \frac{12 - y}{2}$$

$$= 6 - \frac{1}{2}y$$

$\therefore$ Volume

$$= \pi \int_0^4 y \, dy + \pi \int_4^{12} 6 - \frac{1}{2}y \, dy$$

$$= \pi \left[\frac{y^2}{2} \right]_0^4 + \pi \left[6y - \frac{y^2}{4} \right]_4^{12}$$

$$= \pi \left[\frac{16}{2} - 0 \right] + \pi \left[\left(6 \times 12 - \frac{12^2}{4} \right) - \left(6 \times 4 - \frac{4^2}{4} \right) \right]$$

$$= 8\pi + \pi[36 - 20]$$

$$= 8\pi + 16\pi$$

$$= 24\pi \text{ u}^3$$

7. Calculus, EXT1* C3 2006 HSC 4b

$$y = x^2 + 1$$

$$x^2 = y - 1$$

$$V = \pi \int_1^5 x^2 \, dy$$

$$= \pi \int_1^5 y - 1 \, dy$$

$$= \pi \left[\frac{y^2}{2} - y \right]_1^5$$

$$= \pi \left[\left(\frac{25}{2} - 5 \right) - \left(\frac{1}{2} - 1 \right) \right]$$

$$= \pi \left[\frac{15}{2} - \left(-\frac{1}{2} \right) \right]$$

$$= 8\pi \text{ u}^3$$

8. Calculus, EXT1* C3 2007 HSC 9a

$$V = \pi \int_0^1 y^2 \, dx$$

$$= \pi \int_0^1 (x^2 + 1)^2 \, dx$$

$$= \pi \int_0^1 x^4 + 2x^2 + 1 \, dx$$

$$= \pi \left[\frac{1}{5}x^5 + \frac{2}{3}x^3 + x \right]_0^1$$

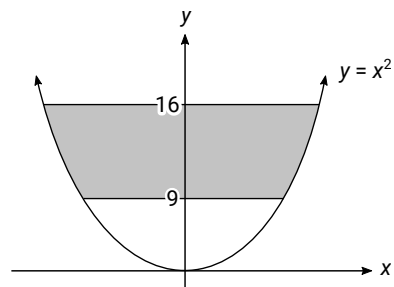
$$= \pi \left[\left(\frac{1}{5} + \frac{2}{3} + 1 \right) - 0 \right]$$

$$= \pi \left[\frac{3}{15} + \frac{10}{15} + 1 \right]$$

$$= \frac{28\pi}{15} \text{ u}^3$$

9. Calculus, EXT1 C3 EQ-Bank 1

i.



ii. Areas either side of y -axis are equal.

$$y = x^2 \Rightarrow x = \sqrt{y}$$

$$\begin{aligned} \therefore A &= 2 \int_9^{16} x \, dy \\ &= 2 \int_9^{16} \sqrt{y} \, dy \\ &= 2 \left[\frac{2}{3} y^{\frac{3}{2}} \right]_9^{16} \\ &= \frac{4}{3} \left[(\sqrt{16})^3 - (\sqrt{9})^3 \right] \\ &= \frac{4}{3} [64 - 27] \\ &= \frac{148}{3} \text{ u}^2 \end{aligned}$$

10. Calculus, EXT1* C3 2019 HSC 13d

$$\begin{aligned} V &= \pi \int_0^1 (x - x^3)^2 dx \\ &= \pi \int_0^1 x^2 - 2x^4 + x^6 dx \\ &= \pi \left[\frac{x^3}{3} - \frac{2}{5} x^5 + \frac{1}{7} x^7 \right]_0^1 \\ &= \pi \left(\frac{1}{3} - \frac{2}{5} + \frac{1}{7} \right) \\ &= \frac{8\pi}{105} \text{ u}^3 \end{aligned}$$

11. Calculus, EXT1 C3 2020 HSC 13b

Find intersection:

$$\sin x = \cos 2x$$

$$\sin x = 1 - 2\sin^2 x$$

$$2\sin^2 x + \sin x - 1 = 0$$

$$(2\sin x - 1)(\sin x + 1) = 0$$

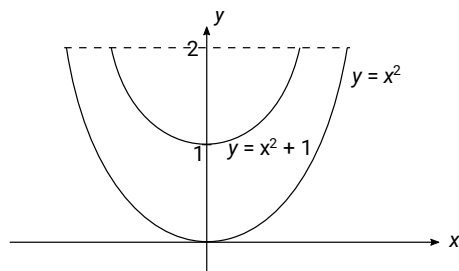
$$\sin x = \frac{1}{2} \quad \text{or} \quad \sin x = -1$$

$$x = \frac{\pi}{6} \quad \quad \quad x = \frac{3\pi}{2}$$

Mean mark 53%.

$$\begin{aligned} V &= \pi \int_0^{\frac{\pi}{6}} (\cos 2x)^2 dx - \pi \int_0^{\frac{\pi}{6}} (\sin x)^2 dx \\ &= \pi \int_0^{\frac{\pi}{6}} \cos^2 2x - \sin^2 x dx \\ &= \pi \int_0^{\frac{\pi}{6}} \frac{1}{2} (1 + \cos 4x) - \frac{1}{2} (1 - \cos 2x) dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{6}} \cos 4x + \cos 2x dx \\ &= \frac{\pi}{2} \left[\frac{1}{4} \sin 4x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{6}} \\ &= \frac{\pi}{8} \left[\sin \frac{2\pi}{3} + 2 \sin \frac{\pi}{3} \right] \\ &= \frac{\pi}{8} \left(\frac{\sqrt{3}}{2} + 2 \times \frac{\sqrt{3}}{2} \right) \\ &= \frac{3\sqrt{3}\pi}{16} \text{ u}^3 \end{aligned}$$

12. Calculus, EXT1 C3 2021 HSC 13a



$$\begin{aligned}
 V &= \pi \int_0^2 y \, dy - \pi \int_1^2 y - 1 \, dy \\
 &= \pi \left[\frac{y^2}{2} \right]_0^2 - \pi \left[\frac{y^2}{2} - y \right]_1^2 \\
 &= \pi(2 - 0) - \pi \left[(2 - 2) - \left(\frac{1}{2} - 1 \right) \right] \\
 &= 2\pi - \pi \left(\frac{1}{2} \right) \\
 &= \frac{3\pi}{2} \text{ u}^3
 \end{aligned}$$

13. Calculus, EXT1 C3 2021 HSC 13c

$$\begin{aligned}
 A &= \text{Area of } \Delta + 2 \left| \int_0^2 1 - \frac{8}{4 + x^2} \, dx \right| \\
 &= \frac{1}{2} \times 4 \times 2 + 2 \left| \left[x - 4 \tan^{-1} \frac{x}{2} \right]_0^2 \right| \\
 &= 4 + 2 \left| (2 - 4 \tan^{-1} 1) - 0 \right| \\
 &= 4 + 2 \left| 2 - \frac{4\pi}{4} \right| \\
 &= 4 + 2(\pi - 2) \\
 &= 2\pi \text{ u}^2
 \end{aligned}$$

14. Calculus, EXT1 C3 2021 SPEC1 4

$$\begin{aligned}
 V_s &= \pi \int_0^\pi \sin^2(x) \, dx \\
 &= \pi \int_0^\pi \frac{1}{2} (1 - \cos(2x)) \, dx \\
 &= \frac{\pi}{2} \int_0^\pi 1 - \cos(2x) \, dx \\
 &= \frac{\pi}{2} \left[x - \frac{1}{2} \sin(2x) \right]_0^\pi \\
 &= \frac{\pi}{2} \left[\pi - \frac{1}{2} \sin(2\pi) - \left(0 - \frac{1}{2} \sin 0 \right) \right] \\
 &= \frac{\pi^2}{2} \text{ u}^3
 \end{aligned}$$

15. Calculus, EXT1 C3 2022 HSC 13b

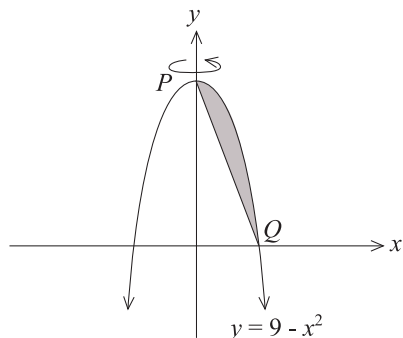
$$\begin{aligned}
 y &= (k + 1) \sin(kx) \\
 V &= \pi \int_0^{\frac{\pi}{2k}} (k + 1)^2 \sin^2(kx) \, dx \\
 &= \pi (k + 1)^2 \int_0^{\frac{\pi}{2k}} \frac{1}{2} [1 - \cos(2kx)] \, dx \\
 &= \left(\frac{\pi}{2} \right) (k + 1)^2 \left[x - \frac{\sin(2kx)}{2k} \right]_0^{\frac{\pi}{2k}} \\
 &= \left(\frac{\pi}{2} \right) (k + 1)^2 \left[\left(\frac{\pi}{2k} - \frac{\sin(\pi)}{2k} \right) - \left(0 - \frac{\sin 0}{2k} \right) \right] \\
 &= \left(\frac{\pi}{2} \right) (k + 1)^2 \left(\frac{\pi}{2k} \right) \\
 &= \frac{\pi^2}{4k} (k + 1)^2
 \end{aligned}$$

Given $V = \pi^2$:

$$\begin{aligned}
 \frac{\pi^2}{4k} (k + 1)^2 &= \pi^2 \\
 (k + 1)^2 &= 4k \\
 k^2 + 2k + 1 &= 4k \\
 k^2 - 2k + 1 &= 0 \\
 (k - 1)^2 &= 0
 \end{aligned}$$

$$\therefore k = 1$$

16. Calculus, EXT1 C3 SM-Bank 2



Equation of PQ :

$$m = -3, \text{ y-intercept} = 9$$

$$y = 9 - 3x$$

Rotating about the y -axis:

$$x_1^2 = 9 - y \quad (\text{parabola})$$

$$y = 9 - 3x_2 \quad (\text{line } PQ)$$

$$3x_2 = 9 - y$$

$$x_2 = 3 - \frac{y}{3}$$

$$x_2^2 = \left(3 - \frac{y}{3}\right)^2$$

$$\begin{aligned} V &= \pi \int_0^9 x_1^2 - x_2^2 \, dy \\ &= \pi \int_0^9 9 - y - \left(3 - \frac{y}{3}\right)^2 \, dy \\ &= \pi \int_0^9 9 - y - \left(9 - 2y + \frac{y^2}{9}\right) \, dy \\ &= \pi \int_0^9 y - \frac{y^2}{9} \, dy \\ &= \pi \left[\frac{y^2}{2} - \frac{y^3}{27} \right]_0^9 \\ &= \pi \left(\frac{81}{2} - 27 \right) \\ &= \frac{27\pi}{2} \text{ units}^3 \end{aligned}$$

17. Calculus, EXT1 C3 SM-Bank 3

$$\begin{aligned} V &= \pi \int_0^1 \frac{1+2x}{1+x^2} \, dx \\ &= \pi \int_0^1 \frac{1}{1+x^2} \, dx + \pi \int_0^1 \frac{2x}{1+x^2} \, dx \\ &= \pi [\tan^{-1}(x)]_0^1 + \pi [\ln(1+x^2)]_0^1 \\ &= \pi (\tan^{-1}1 - \tan^{-1}0) + \pi (\ln 2 - \ln 1) \\ &= \pi \left(\frac{\pi}{4} \right) + \pi (\ln 2) \\ &= \pi \left(\frac{\pi}{4} + \ln 2 \right) \text{ u}^3 \end{aligned}$$

18. Calculus, EXT1* C3 2004 HSC 4c

$$y = 2\sec x$$

$$\therefore y^2 = 4\sec^2 x$$

$$\begin{aligned} V &= \pi \int_0^{\frac{\pi}{3}} y^2 \, dx \\ &= \pi \int_0^{\frac{\pi}{3}} 4\sec^2 x \, dx \\ &= 4\pi \int_0^{\frac{\pi}{3}} \sec^2 x \, dx \\ &= 4\pi [\tan x]_0^{\frac{\pi}{3}} \\ &= 4\pi \left(\tan \frac{\pi}{3} - \tan 0 \right) \\ &= 4\pi (\sqrt{3} - 0) \\ &= 4\sqrt{3} \pi \text{ u}^3 \end{aligned}$$

19. Calculus, EXT1 C3 2013 HSC 12b

$$y = 3\sin \frac{x}{2}$$

$$y^2 = 9\sin^2 \frac{x}{2}$$

Using: $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

$$\begin{aligned}\therefore V &= \pi \int_0^{\frac{3\pi}{2}} 9\sin^2 \frac{x}{2} dx \\ &= \frac{9\pi}{2} \int_0^{\frac{3\pi}{2}} (1 - \cos x) dx \\ &= \frac{9\pi}{2} [x - \sin x]_0^{\frac{3\pi}{2}} \\ &= \frac{9\pi}{2} \left[\left(\frac{3\pi}{2} - \sin \frac{3\pi}{2} \right) - 0 \right] \\ &= \frac{9\pi}{2} \left(\frac{3\pi}{2} + 1 \right) \text{ u}^3\end{aligned}$$

20. Calculus, EXT1* C3 2009 HSC 6a

$$\begin{aligned}V &= \pi \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} y^2 dx \\ &= \pi \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sec^2 x dx \\ &= \pi [\tan x]_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \\ &= \pi \left[\tan \left(\frac{\pi}{3} \right) - \tan \left(-\frac{\pi}{3} \right) \right] \\ &= \pi \left[\sqrt{3} - \left(-\sqrt{3} \right) \right] \\ &= 2\sqrt{3}\pi \text{ u}^3\end{aligned}$$

21. Calculus, EXT1* C3 2012 HSC 14b

$$\begin{aligned}V &= \pi \int_0^1 y^2 dx \\ &= \pi \int_0^1 \left(\frac{3}{(x+2)^2} \right)^2 dx \\ &= \pi \int_0^1 \frac{9}{(x+2)^4} dx \\ &= 9\pi \int_0^1 (x+2)^{-4} dx \\ &= 9\pi \left[-\frac{1}{3}(x+2)^{-3} \right]_0^1 \\ &= 9\pi \left[\left(-\frac{1}{3} \times \frac{1}{3^3} \right) - \left(-\frac{1}{3} \times \frac{1}{2^3} \right) \right] \\ &= 9\pi \left[-\frac{1}{81} + \frac{1}{24} \right] \\ &= 9\pi \left(\frac{19}{648} \right) \\ &= \frac{19\pi}{72} \text{ u}^3\end{aligned}$$

$\therefore$ Volume of the solid is $\frac{19\pi}{72} \text{ u}^3$.

MARKER'S COMMENT: Most students omitted the dx when stating the definite integral in this question!

22. Calculus, EXT1* C3 2014 HSC 14c

$$\begin{aligned}y &= 1 + \sqrt{x} \\ V &= \pi \int_0^4 y^2 dx \\ &= \pi \int_0^4 (1 + \sqrt{x})^2 dx \\ &= \pi \int_0^4 (1 + 2\sqrt{x} + x) dx \\ &= \pi \left[x + \frac{4}{3}x^{\frac{3}{2}} + \frac{1}{2}x^2 \right]_0^4 \\ &= \pi \left[\left(4 + \frac{4}{3} \times 4^{\frac{3}{2}} + \frac{1}{2} \times 4^2 \right) - 0 \right] \\ &= \pi \left(4 + \frac{32}{3} + 8 \right) \\ &= \frac{68\pi}{3} \text{ u}^3\end{aligned}$$

23. Calculus, EXT1* C3 2018 HSC 14b

$$y = x^4 + 1$$

$$x^4 = y - 1$$

$$x^2 = \pm (y-1)^{\frac{1}{2}}$$

$$\text{When } x = 0, \quad y = 1$$

$$\Rightarrow x^2 = (y-1)^{\frac{1}{2}}$$

$$\begin{aligned} \therefore \text{Volume} &= \pi \int_1^{10} x^2 \, dy \\ &= \pi \int_1^{10} (y-1)^{\frac{1}{2}} \, dy \\ &= \pi \times \frac{2}{3} \left[(y-1)^{\frac{3}{2}} \right]_1^{10} \\ &= \frac{2\pi}{3} \left[9^{\frac{3}{2}} - 0 \right] \\ &= \frac{2\pi}{3} (27) \\ &= 18\pi \text{ units}^2 \end{aligned}$$

24. Calculus, EXT1* C3 2008 HSC 6c

$$y = \frac{5}{x-2}$$

$$\begin{aligned} V &= \pi \int_3^6 y^2 \, dx \\ &= \pi \int_3^6 \left(\frac{5}{x-2} \right)^2 \, dx \\ &= 25\pi \int_3^6 \frac{1}{(x-2)^2} \, dx \\ &= 25\pi \left[\frac{-1}{x-2} \right]_3^6 \\ &= 25\pi \left[-\frac{1}{4} - (-1) \right] \\ &= 25\pi \left[\frac{3}{4} \right] \\ &= \frac{75\pi}{4} \text{ u}^3 \end{aligned}$$

25. Calculus, EXT1 C3 2005 HSC 5a

$$y = \sin 2x$$

$$y^2 = \sin^2 2x$$

$$\text{Using: } \sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\begin{aligned} \therefore V &= \pi \int_0^{\frac{\pi}{8}} y^2 \, dx \\ &= \pi \int_0^{\frac{\pi}{8}} \sin^2 2x \, dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{8}} 1 - \cos 4x \, dx \\ &= \frac{\pi}{2} \left[x - \frac{1}{4} \sin 4x \right]_0^{\frac{\pi}{8}} \\ &= \frac{\pi}{2} \left[\left(\frac{\pi}{8} - \frac{1}{4} \sin \frac{\pi}{2} \right) - 0 \right] \\ &= \frac{\pi}{2} \left(\frac{\pi}{8} - \frac{1}{4} \right) \\ &= \frac{\pi}{2} \left(\frac{\pi-2}{8} \right) \\ &= \frac{\pi}{16} (\pi-2) \text{ u}^3 \end{aligned}$$

COMMENT: Michael Wells (1st in state Ext2) would derive this formula in his working from the $\sin^2 x + \cos^2 x = 1$ identity in ~5 seconds *every time* he used it in an exam.

26. Calculus, EXT1* C3 2007 HSC 3a

$$y = \frac{1}{\sqrt{9+x^2}}$$

$$\begin{aligned}\therefore \text{Volume} &= \pi \int_0^3 y^2 \, dx \\ &= \pi \int_0^3 \left(\frac{1}{\sqrt{9+x^2}} \right)^2 \, dx \\ &= \pi \int_0^3 \frac{1}{9+x^2} \, dx \\ &= \pi \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_0^3 \\ &= \pi \left[\frac{1}{3} \tan^{-1} 1 - \frac{1}{3} \tan^{-1} 0 \right] \\ &= \pi \left[\left(\frac{1}{3} \times \frac{\pi}{4} \right) - 0 \right] \\ &= \frac{\pi^2}{12} \text{ u}^3\end{aligned}$$

27. Calculus, EXT1 C3 2014 HSC 12b

$$\begin{aligned}V &= \pi \int_0^{\frac{\pi}{8}} y^2 \, dx \\ &= \pi \int_0^{\frac{\pi}{8}} \cos^2 4x \, dx \\ &= \pi \int_0^{\frac{\pi}{8}} \frac{1}{2} (\cos 8x + 1) \, dx \\ &= \frac{\pi}{2} \left[\frac{1}{8} \sin 8x + x \right]_0^{\frac{\pi}{8}} \\ &= \frac{\pi}{2} \left[\left(\frac{1}{8} \sin \pi + \frac{\pi}{8} \right) - 0 \right] \\ &= \frac{\pi^2}{16} \text{ u}^3\end{aligned}$$

COMMENT: The identities $\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2\cos^2 \theta - 1 = 1 - 2\sin^2 \theta$ are tested every year – know them.

28. Calculus, EXT1 C3 SM-Bank 1

(i) Volume of smaller solid

$$\begin{aligned}&= \pi \int_h^1 \left(\sqrt{1-x^2} \right)^2 \, dx \\ &= \pi \int_h^1 1-x^2 \, dx \\ &= \pi \left[x - \frac{x^3}{3} \right]_h^1 \\ &= \pi \left[\left(1 - \frac{1}{3} \right) - \left(h - \frac{h^3}{3} \right) \right] \\ &= \pi \left(\frac{2}{3} - h + \frac{h^3}{3} \right)\end{aligned}$$

Since smaller solid is $\frac{1}{3}$ volume of sphere,

$$\begin{aligned}\pi \left(\frac{2}{3} - h + \frac{h^3}{3} \right) &= \frac{1}{3} \times \frac{4}{3} \cdot \pi \cdot 1^3 \\ \frac{h^3}{3} - h + \frac{2}{3} &= \frac{4}{9} \\ 3h^3 - 9h + 6 &= 4 \\ \therefore 3h^3 - 9h + 2 &= 0 \quad \dots \text{as required}\end{aligned}$$

29. Calculus, EXT1 C3 2019 SPEC1-N 9

i. Method 1:

$$\begin{aligned}\tan \frac{5\pi}{12} &= \tan \left(\frac{\pi}{4} + \frac{\pi}{6} \right) \\ &= \frac{\tan \frac{\pi}{4} + \tan \frac{\pi}{6}}{1 - \tan \frac{\pi}{4} \cdot \tan \frac{\pi}{6}} \\ &= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} \\ &= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} \\ &= \frac{3 + 2\sqrt{3} + 1}{3 - 1} \\ &= \sqrt{3} + 2\end{aligned}$$

Method 2:

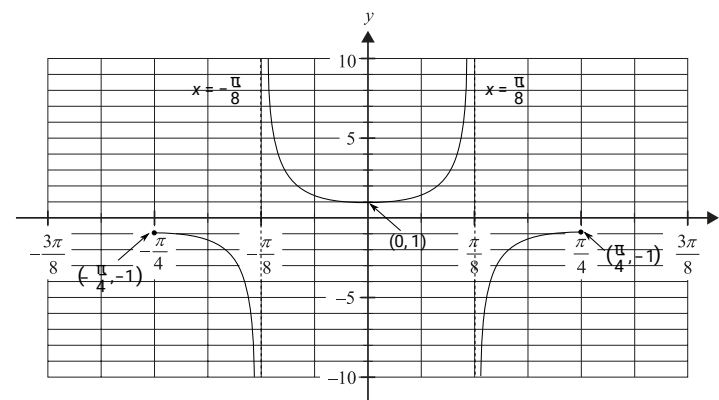
$$\begin{aligned}\tan \frac{5\pi}{6} &= \frac{2\tan \frac{5\pi}{12}}{1 - \tan^2 \frac{5\pi}{12}} \\ -\frac{1}{\sqrt{3}} &= \frac{2\tan \frac{5\pi}{12}}{1 - \tan^2 \frac{5\pi}{12}} \\ -2\sqrt{3}\tan \frac{5\pi}{12} &= 1 - \tan^2 \frac{5\pi}{12} \\ \tan^2 \frac{5\pi}{12} - 2\sqrt{3}\tan \frac{5\pi}{12} - 1 &= 0 \\ \tan \frac{5\pi}{12} &= \frac{2\sqrt{3} \pm \sqrt{12 + 4}}{2} \\ &= \sqrt{3} + 2 \quad (\tan \theta > 0)\end{aligned}$$

ii. Area = $\int_0^{2\sqrt{3}+6} \frac{2}{x^2-4x+8} dx$

$$\begin{aligned}&= \int_0^{2\sqrt{3}+6} \frac{2}{(x-2)^2 + 2^2} \\ &= \left[\tan^{-1} \left(\frac{x-2}{2} \right) \right]_0^{2\sqrt{3}+6} \\ &= \tan^{-1}(\sqrt{3} + 2) - \tan^{-1}(-1) \\ &= \frac{5\pi}{12} - \left(-\frac{\pi}{4} \right) \\ &= \frac{2\pi}{3}\end{aligned}$$

30. Calculus, EXT1 C3 2022 SPEC1 10

a.



♦ Mean mark (a) 49%.

b. $V = \pi \int_{-\frac{\pi}{24}}^{\frac{\pi}{48}} \sec^2(4x) dx$

$$\begin{aligned}&= \frac{\pi}{4} [\tan(4x)]_{-\frac{\pi}{24}}^{\frac{\pi}{48}} \\ &= \frac{\pi}{4} \tan\left(\frac{\pi}{12}\right) - \frac{\pi}{4} \tan\left(-\frac{\pi}{6}\right) \\ &= \frac{\pi}{4} \tan\left(\frac{\pi}{3} - \frac{\pi}{4}\right) - \frac{\pi}{4} \times -\frac{1}{\sqrt{3}} \\ &= \frac{\pi}{4} \left(\frac{\sqrt{3}-1}{1+\sqrt{3}} \times \frac{1-\sqrt{3}}{1-\sqrt{3}} \right) + \frac{\pi}{4\sqrt{3}} \\ &= \frac{\pi}{4} \left(\frac{\sqrt{3}-3-1+\sqrt{3}}{-2} \right) + \frac{\pi}{4\sqrt{3}} \\ &= \frac{\pi}{4} (2 - \sqrt{3}) + \frac{\pi}{4\sqrt{3}} \\ &= \frac{\pi(2\sqrt{3}-3+1)}{4\sqrt{3}} \\ &= \frac{(6-2\sqrt{3})\pi}{12} \\ &= \frac{(3-\sqrt{3})\pi}{6}\end{aligned}$$

♦ Mean mark (b) 55%.

31. Calculus, EXT1 C3 2023 HSC 12e

$$y = \frac{60}{x + 5} \Rightarrow x = \frac{60}{y} - 5$$

◆ Mean mark 49%.

$$\begin{aligned} V &= \pi \int_4^{12} x^2 \, dy + \pi r^2 h \\ &= \pi \int_4^{12} \left(\frac{60}{y} - 5 \right)^2 \, dy + \pi \times 10^2 \times 4 \\ &= 25\pi \int_4^{12} \left(\frac{12}{y} - 1 \right)^2 \, dy + 400\pi \\ &= 25\pi \int_4^{12} \left(\frac{144}{y^2} - \frac{24}{y} + 1 \right) \, dy + 400\pi \\ &= 25\pi \left[\frac{-144}{y} - 24\ln y + y \right]_4^{12} + 400\pi \\ &= 25\pi \left[(-12 - 24\ln 12 + 12) - (-36 - 24\ln 4 + 4) \right] + 400\pi \\ &= 25\pi (24\ln 4 - 24\ln 12 + 32) + 400\pi \\ &= 600\pi \ln (3^{-1}) + 800\pi + 400\pi \\ &= 1200\pi - 600\pi \ln 3 \text{ u}^3 \end{aligned}$$

32. Calculus, EXT1* C3 2013 HSC 15b

Since rotation about the y -axis,

$$(x - 2)^2 = y$$

$$x - 2 = \pm y^{\frac{1}{2}}$$

$$x = 2 \pm y^{\frac{1}{2}}$$

When $x = 0$, $y = 4$

$$\therefore x = 2 - y^{\frac{1}{2}}$$

$$\begin{aligned} \therefore \text{Volume} &= \pi \int_0^4 x^2 \, dy \\ &= \pi \int_0^4 \left(2 - y^{\frac{1}{2}} \right)^2 \, dy \\ &= \pi \int_0^4 \left(4 - 4y^{\frac{1}{2}} + y \right) \, dy \\ &= \pi \left[4y - \left(4 \times \frac{2}{3} \times y^{\frac{3}{2}} \right) + \left(\frac{1}{2} y^2 \right) \right]_0^4 \\ &= \pi \left[4y - \frac{8}{3} y^{\frac{3}{2}} + \frac{1}{2} y^2 \right]_0^4 \\ &= \pi \left[(4 \times 4) - \left(\frac{8}{3} \times 4^{\frac{3}{2}} \right) + \left(\frac{1}{2} \times 4^2 \right) \right] \\ &= \pi \left[16 - \frac{64}{3} + 8 \right] \\ &= \frac{8\pi}{3} \text{ u}^3 \end{aligned}$$

◆ Mean mark 39%.

33. Calculus, EXT1* C3 2015 HSC 16b

◆ Mean mark 38%.

$$y = 8\log_e(x-1)$$

$$\frac{y}{8} = \log_e(x-1)$$

$$e^{\frac{y}{8}} = x-1$$

$$x = e^{\frac{y}{8}} + 1$$

$$\begin{aligned} x^2 &= \left(e^{\frac{y}{8}} + 1\right)^2 \\ &= \left(e^{\frac{y}{8}}\right)^2 + 2e^{\frac{y}{8}} + 1 \\ &= e^{\frac{y}{4}} + 2e^{\frac{y}{8}} + 1 \end{aligned}$$

$$V = \pi \int_0^6 x^2 \, dy$$

$$= \pi \int_0^6 \left(e^{\frac{y}{4}} + 2e^{\frac{y}{8}} + 1\right) \, dy$$

$$= \pi \left[4e^{\frac{y}{4}} + 16e^{\frac{y}{8}} + y \right]_0^6$$

$$= \pi \left[\left(4e^{\frac{6}{4}} + 16e^{\frac{6}{8}} + 6 \right) - (4e^0 + 16e^0 + 0) \right]$$

$$= \pi \left[(4e^{1.5} + 16e^{0.75} + 6) - (4 + 16) \right]$$

$$= \pi [4e^{1.5} + 16e^{0.75} - 14]$$

$$= 118.748\dots$$

$$= 118.7 \text{ (to 1 d.p.)}$$

∴ Volume of the bowl is 118.7 u³.

34. Calculus, EXT1* C3 2016 HSC 15a

Consider C_1 ,

$$V_1 = \pi \int_{-2}^0 y^2 \, dx$$

$$= \pi \int_{-2}^0 4-x^2 \, dx$$

$$= \pi \left[4x - \frac{x^3}{3} \right]_{-2}^0$$

$$= \pi \left[0 - \left(-8 + \frac{8}{3} \right) \right]$$

$$= \frac{16\pi}{3} \text{ u}^3$$

Consider C_2

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$y^2 = 4 - \frac{4x^2}{9}$$

$$V_2 = \pi \int_0^3 4 - \frac{4x^2}{9} \, dx$$

$$= \pi \left[4x - \frac{4x^3}{27} \right]_0^3$$

$$= \pi \left[\left(12 - \frac{4 \cdot 3^3}{27} \right) - 0 \right]$$

$$= 8\pi \text{ u}^3$$

Volume = $V_1 + V_2$

$$= \frac{16\pi}{3} + 8\pi$$

$$= \frac{40\pi}{3} \text{ u}^3$$

35. Calculus, EXT1* C3 2011 HSC 8b

$$\begin{aligned}
 \text{i. } V &= \pi \int_0^h x^2 \, dy \\
 &= \pi \int_0^h y \, dy \\
 &= \pi \left[\frac{1}{2} y^2 \right]_0^h \\
 &= \pi \left(\frac{1}{2} h^2 \right) \\
 &= \frac{\pi h^2}{2} \text{ u}^3
 \end{aligned}$$

$\therefore$ The volume of the paraboloid is $\frac{\pi h^2}{2} \text{ u}^3$

ii. Radius of cylinder (r) = HC

Find x -coordinate of C :

$$\begin{aligned}
 \text{When } y &= h \\
 \Rightarrow x^2 &= h \\
 x &= \sqrt{h} \\
 \therefore r &= \sqrt{h}
 \end{aligned}$$

$$\begin{aligned}
 \text{Volume of cylinder} &= \pi r^2 h \\
 &= \pi \left(\sqrt{h} \right)^2 h \\
 &= \pi h^2
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{ Volume of paraboloid : volume of cylinder} \\
 &= \frac{\pi h^2}{2} : \pi h^2 \\
 &= 1 : 2
 \end{aligned}$$

IMPORTANT: Most common errors:
1-use the correct axis, and 2-check the limits!

♦♦ Mean mark of 24%.

36. Calculus, EXT1* C3 2010 HSC 10b

$$\text{i. Show that } V = \frac{\pi r^3}{3} (2 - 3\sin\theta + \sin^3\theta)$$

$$\sin\theta = \frac{OA}{r}$$

$$\therefore OA = r\sin\theta$$

$$\Rightarrow A = (r\sin\theta, 0), \quad B = (r, 0)$$

$$\begin{aligned}
 \therefore V &= \pi \int_{r\sin\theta}^r y^2 \, dx \\
 &= \pi \int_{r\sin\theta}^r (r^2 - x^2) \, dx \quad (\text{using } x^2 + y^2 = r^2) \\
 &= \pi \left[r^2 x - \frac{x^3}{3} \right]_{r\sin\theta}^r \\
 &= \pi \left[\left(r^3 - \frac{r^3}{3} \right) - \left(r^3 \sin\theta - \frac{r^3 \sin^3\theta}{3} \right) \right] \\
 &= \pi \left(\frac{2r^3}{3} - r^3 \sin\theta + \frac{r^3 \sin^3\theta}{3} \right) \\
 &= \frac{\pi r^3}{3} (2 - 3\sin\theta + \sin^3\theta) \quad \dots \text{ as required}
 \end{aligned}$$

ii. (1) Depth of water remaining = $\frac{1}{2} \times$ original depth:

$$r - r\sin\theta = \frac{1}{2}r$$

$$r(1 - \sin\theta) = \frac{1}{2}r$$

$$1 - \sin\theta = \frac{1}{2}$$

$$\sin\theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{6} \text{ radians}$$

$$\begin{aligned}
 \text{ii. (2) Original Volume} &= \frac{1}{2} \times \frac{4}{3} \pi r^3 \\
 &= \frac{2}{3} \pi r^3
 \end{aligned}$$

$$\begin{aligned}
 \text{New Volume} &= \frac{\pi r^3}{3} \left[2 - 3\sin\left(\frac{\pi}{6}\right) + \sin^3\left(\frac{\pi}{6}\right) \right] \\
 &= \frac{\pi r^3}{3} \left[2 - \left(3 \times \frac{1}{2} \right) + \left(\frac{1}{2} \right)^3 \right]
 \end{aligned}$$

♦♦♦ Mean mark (i) 16%.
MARKER'S COMMENT: A common error was to integrate r^2 to $\frac{1}{3}r^3$ instead of r^2x (note that r is a constant).

♦♦♦ Part (ii) mean marks 3% and 2% for (ii)(1) and (ii)(2) respectively.

MARKER'S COMMENT: Previous parts of a question should always be at the front and centre of a student's mind and direct their strategy.

$$\begin{aligned}
 &= \frac{\pi r^3}{3} \left[2 - \frac{3}{2} + \frac{1}{8} \right] \\
 &= \frac{\pi r^3}{3} \left(\frac{5}{8} \right) \\
 &= \frac{5\pi r^3}{24}
 \end{aligned}$$

$\therefore$ Fraction of original volume left

$$\begin{aligned}
 &= \frac{\frac{5\pi r^3}{24}}{\frac{2}{3}\pi r^3} \\
 &= \frac{5}{24} \times \frac{3}{2} \\
 &= \frac{5}{16}
 \end{aligned}$$